

Yumeng Wang

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Summary

Machine Learning Engineer and Ph.D. researcher with **5+** years of experience building systems across industry and research, including credit risk modeling, scientific machine learning, and large language model agent workflows. Executed end-to-end ML development across data preparation, feature engineering, model design, training, evaluation, and monitoring; coordinated with engineering teams for production deployment. Strong foundation in computational mathematics, statistics, ML, and LLM, with hands-on experience across both research and production environments. Ph.D. candidate in Computational and Applied Mathematics at Missouri University of Science and Technology. Available for full-time roles starting **August 2026**.

Research Experience

ML Research Intern

Jun 2024 – Aug 2024

Lawrence Berkeley National Laboratory (LBNL), Berkeley, CA

► Generative AI for catalyst design

Designed and implemented generative model architectures (VAE, GAN, and diffusion model) for molecular catalyst generation, automating structural exploration beyond the limitations of manual design and exhaustive testing.

Graduate Research Assistant

Jun 2022 – Present

Missouri University of Science and Technology, Rolla, MO

► Scientific surrogate modeling for engineering and applied sciences

Developed scalable surrogate and generative modeling frameworks to accelerate complex scientific simulations across engineering and applied sciences.

• Model reduction for real-time prediction

Built CNN- and attention-based surrogate models to compress high-fidelity solution fields into low-dimensional latent representations, reducing inference time from hours to seconds for real-time digital twin simulation and engineering design optimization.

• Generative models

Developed Wasserstein GAN and VAE-based probabilistic generative models for scientific simulation, transferring generative architectures from image synthesis to computational science; Outperformed deterministic DNN baselines on fluid dynamics and diffusion tasks, with applications in biomedical modeling.

► Transformer-based modeling in scientific computing and marketing analytics

Developed transformer-based learning frameworks to advance scalable modeling across scientific computing, foundation models, and multimodal behavioral analytics.

• Attention-based neural operators

Built an attention-based neural operator leveraging cross-attention and Fourier layers for flexible multi-resolution learning, outperforming FNO on irregular geometries and cross-resolution prediction tasks in scientific and engineering applications.

• Vision-transformer foundation models

Architected and pretrained vision transformer foundation models across 30+ physics model families, leveraging autoregressive learning and equation-aware embeddings to stabilize PINN training and enable data-free downstream prediction.

• Customer attention and behavior analysis in marketing

Built LSTM and multimodal transformer pipelines to integrate spatial, temporal, and audio signals from advertising and behavioral-study videos for customer attention modeling and decision-making analytics.

► LLM-powered finance applications

Developed AI-powered financial applications for quantitative research, signal generation, and LLM-based financial analysis.

• Quantitative Alpha research framework

Built a quantitative alpha research framework on S&P 500 data, implementing factor-based, MLP, transformer, and LLM-integrated signal generation models for systematic strategy development.

- **LLM-powered financial analysis agent**
Developed a ReAct-based LLM financial analysis agent using Google Gemini, enabling dynamic tool orchestration across market data, fundamentals, technical indicators, risk metrics, and news sentiment. Replaced rigid pipelines with Reason–Act–Observe decision-making framework. Implemented a three-layer memory architecture with a RAG retrieval pipeline for semantic search over financial reports and research documents.

Industry Experience

Data scientist Mar 2020 – Apr 2021
Beijing One Zero Dance Technology, Beijing, China

- Drove data-informed credit strategy for Mexico and Vietnam markets, conducting daily SQL analysis to adjust risk thresholds and approval policies in a fast-moving, high-default lending environment, contributing to overall profit margin growth.
- Built a Python automation tool connecting to SQL to generate weekly business metric reports, eliminating manual data pulls for a team of 10+ analysts.

Machine learning engineer, FinSight Lab Mar 2019 – Feb 2020
Beijing Fantaike Technology Co., Ltd., Beijing, China

- Built end-to-end ML pipeline for automated loan approval, including data cleaning, feature engineering, model training (XGBoost/LR), and monthly model updates and monitoring; coordinated with engineers for production deployment and validated online-offline consistency; auto-approved segment achieved lower default rate than overall default rate, reducing manual review workload.
- Developed a time series model to forecast daily loan application volume, enabling data-driven staffing allocation.
- Built a customer churn prediction model for a commercial bank, working onsite for 2 months with SQL-based feature engineering and LightGBM.

Data science intern Apr 2018 – Sep 2018
DiDi Chuxing Technology Co., Ltd., Beijing, China

- Wrote 100+ SQL queries on Hadoop to monitor core funnel metrics (order placement, driver acceptance, trip completion, retention), directly informing product decisions during expansion from 2 to 10+ cities.
- Built Tableau dashboards connected to MySQL to centralize daily KPI tracking across cities, serving as the primary reporting tool for the entire business line.

Education

Ph.D. Candidate, Computational and Applied Mathematics Aug 2021 – Present
Missouri University of Science and Technology, Rolla, MO

- Research areas: Scientific Machine Learning and LLM-powered Applications
- Award: Eloë Graduate Research Award, 2024

M.S., Management Science and Engineering Sep 2016 – Mar 2019
Tongji University, Shanghai, China

B.S., Engineering Management Sep 2011 – Jul 2015
Beijing University of Civil Engineering and Architecture, Beijing, China

Technical Skills

Programming	Python, MATLAB, C, SQL, Git, Linux
ML Engineering	PyTorch, TensorFlow, Sklearn, XGBoost, LightGBM, Docker, AWS, HPC, Weights & Biases
LLM & Agentic AI	LangChain, RAG, Prompt engineering, Tool use, Agent workflows (ReAct), LLM APIs, HuggingFace, LoRA fine-tuning
Scientific ML	Foundation models, Neural operators, PINNs, GANs, VAEs, Diffusion models, Transformer-based models
Data & Analytics	NumPy, Pandas, SciPy, Tableau, A/B Testing
Maths & Stats	Numerical analysis, Numerical algorithms, Optimization, Matrix theory, Stochastic processes, Econometrics, Mathematical statistics